Heer Kumari Perchani

(825) 889-6236 | perchani@ualberta.ca | heer.perchani@gmail.com | linkedin.com | github.com | Edmonton, AB
Authorized to Work in Canada (Valid Work Permit)

Education

University of Alberta Expected May 2027
B.S. in Computing Science, Minor in Statistics *Edmonton, AB*
Awards: Dean's Honor Roll, First Class Standing
Coursework: Computer Vision, NLP, Machine Learning, Reinforcement Learning, Data Structures & Algorithms, Probability & Statistics, Stochastic Processes
Certifications: Google Data Analytics Specialization (Coursera, 2024)

Technical Skills

Languages: Python, SQL, R, Java, C/C++, JavaScript, Kotlin, Bash, HTML/CSS, Regex
ML & AI: PyTorch, Scikit-Learn, Hugging Face, OpenCV, Gemini API, OpenAI API, Gym/Panda-Gym
Data Analysis & BI: NumPy, Pandas, Matplotlib, Seaborn, Power BI, Tableau, Power Query, DAX, IBM SPSS, Excel
Data Engineering: SQL Server, ETL Pipelines, Firebase
Tools & Practices: Git/GitHub, VS Code, Jupyter, Android Studio, Flask, Agile, Scrum, TDD

Experience

Machine Learning Fellow May 2026 – Present
AI4Good Lab & Amii *Edmonton, AB*
- Delivered 4 ML projects for Kaggle competitions, architecting CNNs for image classification tasks
- Engineered ResNeXt-50 and EfficientNetV2 classifiers for cassava leaf disease detection, achieving 87.38% accuracy
- Standardized 8 real-world datasets (80,000+ records) and partnered with Humane Canada and 5 shelters, including 3 of Canada's largest, to source domain expertise for the model

Community Caretaker Sep 2025 – Apr 2026
St. Joseph's College *Edmonton, AB*
- Resolve 10+ resident service requests weekly, supporting 30+ residents daily and 250+ building-wide
- Spearhead 1 floor event monthly and co-organize 5+ residence-wide events, driving community engagement

Teaching Assistant – CMPUT 175 Sep 2025 – Dec 2025
University of Alberta *Edmonton, AB*
- Facilitated technical labs and debugged Python codebases for 130+ students each term
- Assessed 20+ assignments weekly, delivering targeted feedback on code architecture and logic

Projects

FurCast: Predictive Animal Shelter Analytics | *Python, Scikit-Learn, Gemini API, Git* GitHub
- Architected a predictive model for shelter length-of-stay and outcomes, projected to cut admin workload by 10+ hours weekly across Edmonton shelters
- Engineered a nested Random Forest ensemble, isolating species-specific models for cats and dogs, and unified shelter records with external demographic data via custom ETL pipelines
- Deployed the Gemini API to impute missing breed attributes, lifting overall model accuracy

Netflix Movies & TV Shows Dashboard | *Power BI, Power Query, DAX* GitHub
- Processed and enriched 8,278 Netflix titles in Power Query, resolving missing values and applying sentiment tags
- Designed a Power BI dashboard with KPI visualizations, surfacing the post-2015 surge in TV show output

Human Emotion Classification Web App | *PyTorch, Flask, Python* GitHub
- Developed a full-stack emotion-detection application with PyTorch and Flask, classifying facial expressions into 5 categories in real time
- Architected the image-processing and CNN training pipeline end-to-end, achieving 71% accuracy

Feelin': Cloud-Connected Android App | *Java, Firebase, Google Maps API, Git* GitHub
- Co-engineered an Android mood-tracking app, implementing Firebase for real-time, synchronized data storage
- Integrated the Google Maps API and ran usability tests with 25 users, refining the UI to reduce input friction

Leadership & Activities

Director of Events and Marketing Aug 2024 – Oct 2025
Interdepartmental Science Students' Society *Edmonton, AB*
- Produced 25+ promotional posters using Canva and Adobe, sustaining consistent visual branding across events
- Boosted event visibility and engagement through strategic content planning
- Ran 20+ office hours and partnered with executive teams to align marketing strategy with organizational goals